

Multiple Choice (10 marks)

1. Considering 0.1 M aqueous solutions of each of the following, which solution has the lowest pH?
 - (a) Na_2CO_3
 - (b) Na_3PO_4
 - (c) Na_2S
 - (d) NaCl
2. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
 - (a) 23.56 GHz
 - (b) 42.58 MHz
 - (c) 74.34 kHz
 - (d) 13.93 MHz
3. Of the following solutions, which will have the highest ionic strength? (Assume complete dissociation.)
 - (a) 0.050 M AlCl_3
 - (b) 0.100 M NaCl
 - (c) 0.050 M CaCl_2
 - (d) 0.100 M HCl
4. Of the following ionic substances, which has the greatest lattice enthalpy?
 - (a) MgO
 - (b) MgS
 - (c) NaF
 - (d) NaCl
5. The ^{13}C spectrum of which isomer of C_6H_{14} has lines with three distinct chemical shifts?
 - (a) hexane
 - (b) 2-methylpentane
 - (c) 3-methylpentane
 - (d) 2,3-dimethylbutane

True / False (10 marks)

Answer True or False

6. True or False: Carbonic acid is not generally considered a toxic pollutant in water due to its natural occurrence and role in buffering systems.
7. True or False: HCl and NaOH are classified as a conjugate acid-base pair in acid-base chemistry.
8. True or False: Doubling the volume of a buffer solution significantly increases its pH due to dilution effects.
9. True or False: A maximum of three phases can be in equilibrium in a three-component mixture under any conditions.
10. True or False: Unpaired electrons are essential for both paramagnetism and ferromagnetism in materials.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of protein absorption at a wavelength of 190 nm and explain how this relates to $\rightarrow \pi^*$ excitations within the ultraviolet region of the electromagnetic spectrum.
12. Discuss the factors that contribute to the differences in relaxation times T_1 and T_2 in nuclear magnetic resonance, emphasizing the role of molecular motion at the Larmor frequency.

End of Paper